IGCSE 0607 Practice Questions — E4.5 — Combined Questions & Markscheme

Parallel lines

Total Questions: 18 — **Total Marks:** 110 Marks

Table of Contents & Quick Navigation

- **E4.5** Parallel lines
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

1 Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

2 Line L has the equation $4x + 2y = 9$.
Find the equation of the line parallel to L that passes through the point $(3, -1)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L has the equation $y = 6 - 3x$.

Find the equation of the line parallel to L that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 Line L_1 has the equation $y = 3x - 7$.

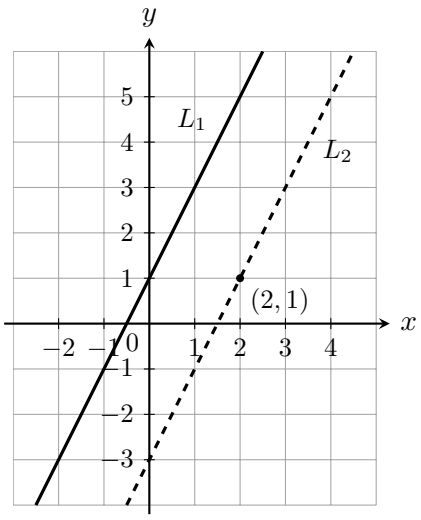
(a) Write down the gradient of a line that is parallel to L_1.

Gradient = ... **[1]**

(b) Line L_2 is parallel to L_1 and passes through the point $(2, -5)$.
Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

y = ... **[3]**

5 The diagram shows two parallel straight lines, L_1 and L_2, on a coordinate grid.
Line L_1 has equation $y = 2x + 1$ and line L_2 passes through the point $(2, 1)$.



(a) Write down the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2 in the form $y = mx + c$.

$y =$ ..... **[2]**

(c) A third line L_3 is parallel to L_1 and passes through the point $(-1, 5)$.
Find the equation of line L_3 in the form $y = mx + c$.

$y =$ ..... **[2]**

6 A straight line L_1 has the equation $2y - 5x = 8$.

(a) Find the gradient of line L_1.

..... **[2]**

(b) Line L_2 is parallel to L_1 and passes through the point $(0, -3)$.
Write down the equation of L_2.

$y =$ ..... **[2]**

(c) Line L_3 is parallel to L_1 and passes through the point $(4, 1)$.
Find the equation of L_3 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

7 The points A and B have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line L is parallel to the line segment AB and passes through the point $(4, -2)$.

(a) Find the gradient of the line segment AB.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

8 Line A has the equation $3x + 2y = 14$.

(a) Find the gradient of line A.

Gradient = ... **[2]**

(b) Line B is parallel to line A and passes through the point $(-4, 9)$.
Find the equation of line B.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[4]**

9 Line L_1 has equation $3x + 4y = 12$.

(a) Find the gradient of line L_1.

..... **[2]**

(b) Line L_2 is parallel to line L_1 and passes through the point $(4, 3)$.
Find the equation of line L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(c) Line L_2 crosses the x-axis at A and the y-axis at B.
Find the coordinates of the midpoint of AB.

(..... , ...) **[3]**

10 Line P has equation $4x - ky = 8$.
Line Q has equation $y = (k-3)x + 2$.
The lines P and Q are parallel.

(a) Find the two possible values of the constant k.

$k =$ ..... **[4]**

(b) Given that line P has a positive gradient,

(i) find the equation of line P in the form $y = mx + c$.

$y =$ ..... **[2]**

(ii) A line R is parallel to line P and passes through the point $(-5, 7)$.
Find the equation of line R in the form $ax + by = c$, where a, b, and c are integers.

..... **[2]**

11 Line M has the equation $5x - 2y = 15$.
Point P has coordinates $(-4, 1)$.
Find the equation of the line parallel to M that passes through P.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

12 A is the point $(-3, 2)$ and B is the point $(5, 6)$.

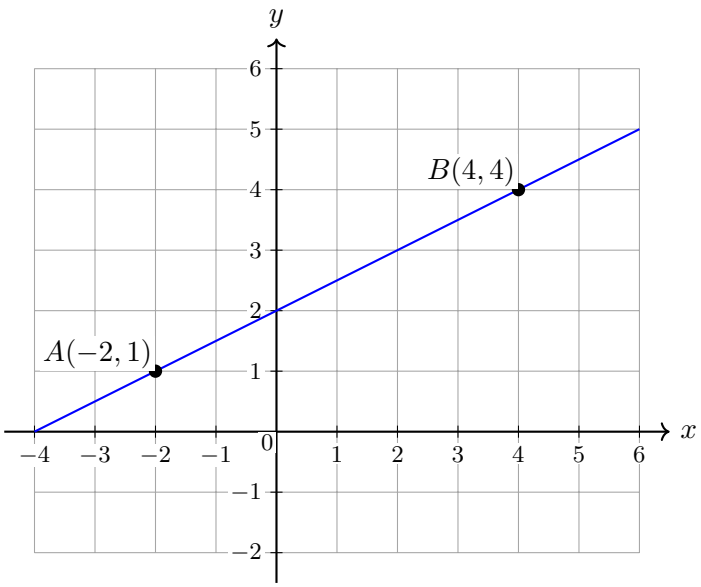
(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Line L is parallel to AB and passes through the point $(4, -1)$.
Find the equation of line L, giving your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

13 The diagram shows the points $A(-2, 1)$ and $B(4, 4)$ joined by a straight line.



(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Line M passes through the point $(2, -3)$ and is parallel to the line AB.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line M intersects the x-axis at point P and the y-axis at point Q.
Calculate the length of the line segment PQ.
Give your answer correct to 3 significant figures.

..... **[4]**

14 Points $A(-1, 5)$ and $B(3, -3)$ lie on line L_1.
Line L_2 is parallel to L_1 and passes through the point $C(2, 7)$.
Line L_2 crosses the x-axis at D and the y-axis at E.

(a) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) Calculate the exact area of triangle ODE, where O is the origin $(0, 0)$.

..... $units^2$ **[3]**

15 The points $P(-2, 3)$, $Q(4, 15)$ and $R(k, 2k + 1)$ lie on a coordinate plane, where k is a constant.

(a) Line L passes through R and is parallel to the line segment PQ.
Given that the y-intercept of line L is -5, find the value of k.

$k =$ ..... **[4]**

(b) Find the equation of the line passing through $Q(4, 15)$ that is parallel to the line $5x - 2y + 9 = 0$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

16 Line L_1 has the equation $3x - 4y = 12$.
Line L_2 is parallel to L_1.

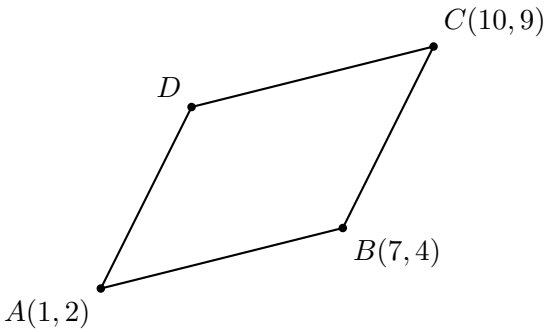
Line L_2 crosses the x-axis at point A and the y-axis at point B.
The length of the line segment AB is 15.

Find the two possible equations of L_2.
Give your answers in the form $y = mx + c$.

.....

..... **[7]**

17 The points $A(1,2)$, $B(7,4)$, and $C(10,9)$ are three vertices of a parallelogram $ABCD$.
Line CD is parallel to line AB and line AD is parallel to line BC.



Not to scale

(a) Find the equation of the line CD.

..... **[3]**

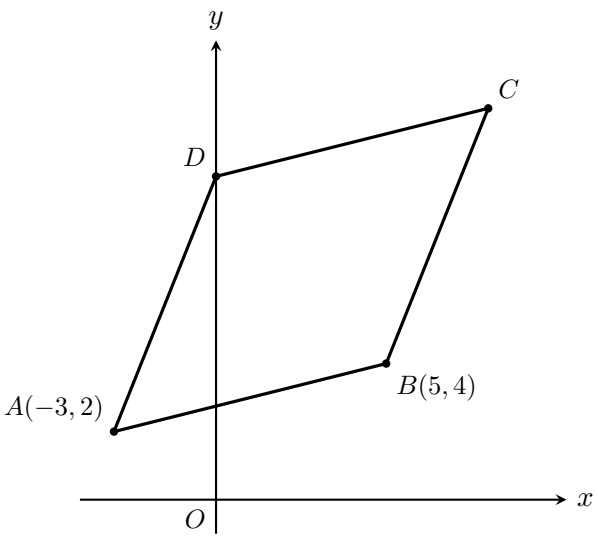
(b) Find the equation of the line AD.

..... **[3]**

(c) Work out the coordinates of the vertex D.

(..... , ...) **[3]**

18 The diagram shows a sketch of a parallelogram $ABCD$ on a coordinate grid.
Vertex A has coordinates $(-3, 2)$ and vertex B has coordinates $(5, 4)$.
The line passing through B and C has equation $5x - 2y = 17$.



(a) Find the equation of the line passing through A and D in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Vertex D lies on the y-axis.
Write down the coordinates of D and find the coordinates of vertex C.

$D =$ ..... , $C =$ ..... **[3]**

(c) Line L is parallel to AB and passes through the midpoint of BC.
Find the equation of line L in the form $ax + by = c$, where a, b, and c are integers.

..... **[4]**

(d) Calculate the length of diagonal AC, giving your answer correct to 3 significant figures.

$AC =$ ..... **[2]**

1 Identifies the gradient $m = -2$ and sets up $4 = -2(3) + c$ or $y - 4 = -2(x - 3)$ **[M1]**

$-2x + 10$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Rearranges equation to find gradient, e.g., $2y = -4x + 9$ implies $m = -2$ **[M1]**

Substitutes $(3, -1)$ into $y = -2x + c$ or writes $y - (-1) = -2(x - 3)$ **[M1]**

$-2x + 5$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 Gradient $m = -3$ **[B1]**

$5 = -3(-2) + c$ or $y - 5 = -3(x - (-2))$ **[M1]**

$y = -3x - 1$ **[A1]**

[Back to Question] *[Total Marks: 3]*

4 (a) 3 **[B1]**

(b) Uses $m = 3$ in $y = 3x + c$ **[M1]**

Substitutes $(2, -5)$: $-5 = 3(2) + c$ **[M1]**

$3x - 11$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) 2 **[B1]**

(b) $1 = 2(2) + c$ or $y - 1 = 2(x - 2)$ **[M1]**

$2x - 3$ **[A1]**

(c) $5 = 2(-1) + c$ or $y - 5 = 2(x + 1)$ **[M1]**

$2x + 7$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 (a) Rearranges the equation to make y the subject: $2y = 5x + 8$ or $y = \frac{5}{2}x + 4$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

(b) Identifies the gradient of L_2 is $\frac{5}{2}$ and the y-intercept is -3 **[M1]**

$\frac{5}{2}x - 3$ (or $2.5x - 3$) **[A1]**

(c) Substitutes $(4, 1)$ and $m = \frac{5}{2}$ into linear equation: $y - 1 = \frac{5}{2}(x - 4)$ or $1 = \frac{5}{2}(4) + c$ **[M1]**

Eliminates the fraction: $2y - 2 = 5(x - 4)$ or $2y - 2 = 5x - 20$ **[M1]**

$5x - 2y = 18$ (or $-5x + 2y = -18$) **[A1]**

[Back to Question] *[Total Marks: 7]*

7 **(a)** $\dfrac{-1-5}{5-(-3)}$ or $\dfrac{5-(-1)}{-3-5}$ **[M1]**

$-\dfrac{3}{4}$ **[A1]**

(b) $m = -\dfrac{3}{4}$ **[M1]**

$-2 = -\dfrac{3}{4}(4) + c$ or $y - (-2) = -\dfrac{3}{4}(x - 4)$ **[M1]**

$y = -\dfrac{3}{4}x + 1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** Rearranges equation to $2y = -3x + 14$ or $y = -\dfrac{3}{2}x + 7$ **[M1]**

$-\dfrac{3}{2}$ or -1.5 **[A1]**

(b) Uses gradient $-\dfrac{3}{2}$ with point $(-4, 9)$: $9 = -\dfrac{3}{2}(-4) + c$ **[M1]**

Finds $c = 3$, giving $y = -\dfrac{3}{2}x + 3$ **[A1]**

Multiplies through by 2 and rearranges: $3x + 2y = 6$ **[M1]**

$3x + 2y = 6$ **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** $4y = -3x + 12$ or $y = -\dfrac{3}{4}x + 3$ **[M1]**

$-\dfrac{3}{4}$ (or -0.75) **[A1]**

(b) Gradient of $L_2 = -\dfrac{3}{4}$ **[B1]**

$y - 3 = -\dfrac{3}{4}(x - 4)$ or $3(4) + 4(3) = c$ **[M1]**

$3x + 4y = 24$ **[A1]**

(c) $A = (8, 0)$ **[B1]**

$B = (0, 6)$ **[B1]**

$(4, 3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** Finds the gradient of line P in terms of k: $m_P = \frac{4}{k}$ **[B1]**

Sets gradients equal: $\frac{4}{k} = k - 3$ **[M1]**

Forms a 3-term quadratic equation: $k^2 - 3k - 4 = 0$ **[M1]**

4 and -1 **[A1]**

(b) **(i)** Identifies that $k = 4$ produces a positive gradient ($m = 1$) while $k = -1$ produces a negative gradient ($m = -4$) **[M1]**

$x - 2$ **[A1]**

(ii) Substitutes $(-5, 7)$ and $m = 1$ into equation: $y - 7 = 1(x - (-5))$ **[M1]**

$x - y = -12$ or $y - x = 12$ (or equivalent integer form) **[A1]**

[Back to Question] *[Total Marks: 8]*

11 Rearranges $5x - 2y = 15$ to find the gradient, e.g. $2y = 5x - 15$

y $= \frac{5}{2}x - 7.5$, so $m = \frac{5}{2}$ [M1]

Substitutes $(-4, 1)$ and $m = \frac{5}{2}$ into $y = mx + c$: $1 = \frac{5}{2}(-4) + c$ **[M1]**

$\frac{5}{2}x + 11$ or $2.5x + 11$ **[A1]**

[Back to Question] *[Total Marks: 3]*

12 **(a)** $\frac{6 - 2}{5 - (-3)}$ oe **[M1]**

$\frac{1}{2}$ or 0.5 **[A1]**

(b) $y - (-1) = \frac{1}{2}(x - 4)$ oe **[M1]**

$2(y + 1) = x - 4$ or $2y + 2 = x - 4$ **[M1]**

$x - 2y = 6$ or $-x + 2y = -6$ **[A1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** Uses the gradient formula: $\frac{4 - 1}{4 - (-2)}$ **[M1]**

$\frac{3}{6} = \frac{1}{2}$ or 0.5 **[A1]**

(b) States or implies gradient $m = 0.5$ **[B1]**

Substitutes $(2, -3)$ into $y = 0.5x + c$: $-3 = 0.5(2) + c$ **[M1]**

$0.5x - 4$ (or $\frac{1}{2}x - 4$) **[A1]**

(c) Finds coordinates of P by setting $y = 0$: $0 = 0.5x - 4 \implies x = 8$, so $P(8, 0)$ **[B1]**

Finds coordinates of Q by setting $x = 0$: $y = -4$, so $Q(0, -4)$ **[B1]**

Applies distance formula: $\sqrt{(8-0)^2 + (0-(-4))^2}$ or $\sqrt{8^2 + (-4)^2}$ **[M1]**

8.94 (from $\sqrt{80}$) **[A1]**

[Back to Question] *[Total Marks: 9]*

14 (a) Calculates the gradient of L_1: $m = \dfrac{-3-5}{3-(-1)}$ **[M1]**

$m = \dfrac{-8}{4} = -2$ **[A1]**

Uses parallel gradient -2 and point $C(2, 7)$ to find c: $7 = -2(2) + c$ **[M1]**

$-2x + 11$ **[A1]**

(b) Finds y-intercept $E(0, 11)$ and sets $y = 0$ to find x-intercept D: $0 = -2x + 11$

x = 5.5[$M1$]

Sets up area calculation for right-angled triangle: $\dfrac{1}{2} \times 5.5 \times 11$ **[M1]**

30.25 or $\dfrac{121}{4}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 (a) Calculates gradient of PQ: $\dfrac{15-3}{4-(-2)} = 2$ **[B1]**

Writes equation of line L as $y = 2x - 5$ **[M1]**

Substitutes $R(k, 2k+1)$ into $y = 2x - 5$ or equates gradient OR' with 2:
$\dfrac{(2k+1)-(-5)}{k-0} = 2$ **[M1]**

Wait: $2k + 1 = 2k - 5$ has no solution — adjust $R(k, 3k-2)$: $3k - 2 = 2k - 5$

k = -3[$A1$]

(b) Finds gradient of given line: $2y = 5x + 9$

m = 2.5($or\dfrac{5}{2}$) [B1]

Substitutes $(4, 15)$: $15 = 2.5(4) + c$

c = 5[$M1$]

$2.5x + 5$ or $\dfrac{5}{2}x + 5$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 Gradient of $L_1 = \dfrac{3}{4}$ oe **[B1]**

Equation of L_2 is $y = \dfrac{3}{4}x + c$ or $3x - 4y = k$ **[M1]**

x-intercept of L_2 is $-\frac{4c}{3}$ (or $\frac{k}{3}$) AND y-intercept is c (or $-\frac{k}{4}$) oe **[M1]**

$(-\frac{4c}{3})^2 + c^2 = 15^2$ or $(\frac{k}{3})^2 + (-\frac{k}{4})^2 = 15^2$ oe **[M1]**

$\frac{25c^2}{9} = 225$ or $\frac{25k^2}{144} = 225$ **[M1]**

$c = \pm 9$ or $k = \pm 36$ **[A1]**

$y = \frac{3}{4}x + 9$ and $y = \frac{3}{4}x - 9$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Gradient of $AB = \frac{4-2}{7-1} = \frac{1}{3}$ **[M1]**

Substitute $(10, 9)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 9 = \frac{1}{3}(x - 10)$ **[M1]**

$y = \frac{1}{3}x + \frac{17}{3}$ or $3y = x + 17$ **[A1]**

(b) Gradient of $BC = \frac{9-4}{10-7} = \frac{5}{3}$ **[M1]**

Substitute $(1, 2)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 2 = \frac{5}{3}(x - 1)$ **[M1]**

$y = \frac{5}{3}x + \frac{1}{3}$ or $3y = 5x + 1$ **[A1]**

(c) Equate equations from (a) and (b): $\frac{1}{3}x + \frac{17}{3} = \frac{5}{3}x + \frac{1}{3}$ **[M1]**

$4x = 16 \implies x = 4$ **[A1]**

$y = 7$ **[A1]**

[Back to Question] *[Total Marks: 9]*

18 **(a)** Gradient of $BC = 2.5$ (or $\frac{5}{2}$) **[B1]**

$2 = 2.5(-3) + c$ or $y - 2 = 2.5(x + 3)$ **[M1]**

$2.5x + 9.5$ (or $\frac{5}{2}x + \frac{19}{2}$) **[A1]**

(b) $D = (0, 9.5)$ **[B1]**

Translation $\overrightarrow{AB} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}$ applied to D **[M1]**

$C = (8, 11.5)$ **[A1]**

(c) Gradient of $AB = \frac{4-2}{5-(-3)} = \frac{1}{4}$ **[B1]**

Midpoint of $BC = \left(\frac{5+8}{2}, \frac{4+11.5}{2}\right) = (6.5, 7.75)$ **[B1]**

$y - 7.75 = \frac{1}{4}(x - 6.5)$ **[M1]**

$2x - 8y = -49$ (or $-2x + 8y = 49$) **[A1]**

(d) $\sqrt{(8-(-3))^2+(11.5-2)^2}$ $(=\sqrt{211.25})$ **[M1]**

14.5 **[A1]**

[Back to Question] *[Total Marks: 12]*